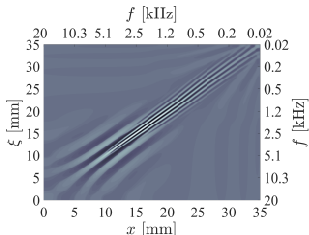
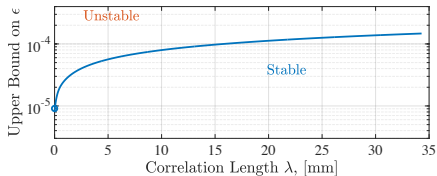
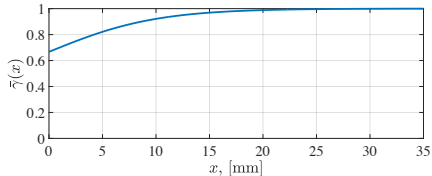
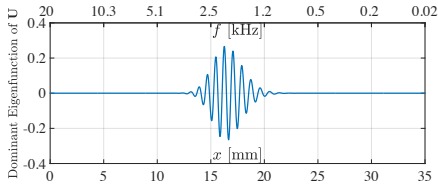


$$\gamma(x, t) = \bar{\gamma}(x) + \tilde{\gamma}(x, t) \quad \text{Covariance: } \mathbb{E}[\tilde{\gamma}(x, t)\tilde{\gamma}(\xi, \tau)] = \frac{\epsilon^2}{\lambda\sqrt{2\pi}} e^{-\frac{(x-\xi)^2}{2\lambda^2}} \delta(t - \tau)$$



$U(x, \xi)$: worst case covariance
of membrane displacement



1st eigenfunction of $U(x, \xi)$
(1st term in K-L expansion)